# MCO-1D

## Operator Manual

How to fly, talk to, and work with your ornithopter drone.

| | |
|---|---|
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| **Audience** | Pilot / Operator (no engineering background required) |
| **Companion** | MCO-1D Commissioning Manual v20 (hardware and build) |

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# 1 What This Manual Is For

This manual tells you — the pilot — everything you need to operate the MCO-1D ornithopter drone. It assumes the aircraft has already been built and commissioned. You do not need to know anything about the software, hardware, or engineering.

## What this manual covers:

- How to start up the ground station on your laptop
- How to talk to the aircraft using plain English conversation
- How to fly — arming, hovering, cruising, landing
- What every position on the throttle lever does
- How the braking system works and when to use it
- How the aircraft learns to fly better over time, and how to help it
- How to run automated practice flight programmes with minimal intervention
- The special procedure for altitude ceiling exploration (airspace-gated)
- After-flight shutdown and how to preserve the aircraft's learning
- Troubleshooting common problems

◆ **NOTE** This manual contains no Python code, no command names, and no technical jargon. Everything is in plain English. For engineering detail see the Commissioning Manual.

## What this manual does not cover:

- Building the aircraft (Wing Manufacturing Manual)
- Commissioning and first-flight test cards (Commissioning Manual)
- Hardware maintenance (Commissioning Manual)
- Regulatory compliance — consult your aviation authority before any flight

## The Aircraft at a Glance

| | |
|---|---|
| **Aircraft** | MCO-1D metachronal cascade ornithopter SAR drone |
| **Wings** | 8 — four tandem pairs, each independently controlled |
| **Weight** | ~4.1–5.1 kg assembled and flying |
| **Stall speed** | 4.85 m/s (~9.4 knots) |
| **Maximum speed** | 28 m/s (~54 knots) |
| **Flapping limit** | 3.0 Hz — hard limit, cannot be overridden |
| **Flight computer** | Raspberry Pi on the aircraft — talks to your laptop |
| **Ground station** | Your laptop — connects to the aircraft over Wi-Fi |
| **Pilot interface** | Plain English conversation via the AI assistant |

# 2 Equipment You Will Need

Before your first flight session, make sure you have everything below. The aircraft side is prepared by whoever commissioned the drone.

## Aircraft side (pre-commissioned)

- MCO-1D assembled and commissioned per the Commissioning Manual
- Charged battery packs (one minimum; three recommended for a learning session)
- Clear open flying area — minimum 20×20 m flat ground, no overhead obstructions
- Landing pad (1×1 m flat surface, clearly marked)

## Ground station side (your laptop)

- **Laptop or tablet** — Windows, macOS, or Linux. Any modern browser.
- **Python 3.9 or later** — installed on the laptop.
- **Two Python packages:** install once with: `pip install aiohttp anthropic`
- **Anthropic API key** — from console.anthropic.com. Set as an environment variable.
- **Three files from the MCO-1D package** (same folder on your laptop):
  `mco_ws_bridge.py` — aircraft link
  `mco_api_proxy.py` — AI link
  `mco_ground_station_launcher.sh` — starts everything
- **Ground station interface** — `mco_ai_ground_station.jsx` in Claude Artifacts or a local server

■ **CAUTION** Your Anthropic API key is a personal credential. Never share it or put it in a file. Always keep it in an environment variable.

## Network setup

The aircraft's Raspberry Pi creates a Wi-Fi network or connects to your router. Your laptop connects to the same network. The default aircraft IP is `192.168.7.1`. If yours differs, whoever built the aircraft will tell you.

◆ **NOTE** If the aircraft IP is not 192.168.7.1, use the correct address in the launch command in Section 3.

# 3 Starting the Ground Station

The ground station runs on your laptop. Three components must all be running before you open the interface. The launcher handles two of them automatically.

## What the three parts do

| | |
|---|---|
| Aircraft link (mco_ws_bridge.py) | Connects your browser to the aircraft over the local network. All commands travel through here; all a |
| AI link (mco_api_proxy.py) | Connects your browser to Claude. Your plain English goes in, aircraft commands come out. The AI r |
| Browser interface (mco_ai_ground_station.jsx) | The conversation window. Shows live flight instruments at the top. Your words on the right, AI replies |

## Starting up — step by step

**1** **Power on the aircraft**

Connect the battery. Wait for the aircraft's status light to show solid green (usually 10–15 seconds). The Raspberry Pi is booting and starting the flight software.

Do not try to arm or fly until the light is solid green.

**2** **Connect your laptop to the aircraft's network**

Join the Wi-Fi network. Confirm you can reach the aircraft: open a terminal and type `ping 192.168.7.1`. You should see replies.

Replace 192.168.7.1 with your aircraft's actual IP address if different.

**3** **Set your Anthropic API key**

Open a terminal. Type the following, replacing the text after = with your actual key:

```
export ANTHROPIC_API_KEY=sk-ant-...
```

Do this in the same terminal window you will use in Step 4. Not saved between sessions.

**4** **Run the launcher**

In the same terminal, go to the folder with the MCO-1D ground station files and run:

```
bash mco_ground_station_launcher.sh 192.168.7.1
```

You should see two confirmation messages. The terminal then waits — leave it running.

**5** **Open the browser interface**

Go to claude.ai, start a conversation, attach mco_ai_ground_station.jsx, and ask Claude to render it as an artifact. The ground station appears.

**6** **Check the connection badge**

Top-right corner: green DISARMED = connected and ready. Red OFFLINE = not connected. Check the launcher and aircraft power if OFFLINE.

ARMED (amber) means armed and about to fly.

■ **CAUTION** If OFFLINE persists after 30 seconds: (1) check the aircraft IP, (2) confirm the aircraft light is solid green, (3) restart the launcher.

# 4 The Conversation Interface

The ground station is a conversation. You type plain English. The AI understands your intent, operates the aircraft, and tells you what happened. You never need to know a single command name.

## What you see on screen

| | |
|---|---|
| **Flight instruments (top)** | Six round analogue gauges: airspeed, altitude, vertical speed, power, battery, and flight state. Upda |
| **Connection badge (top right)** | OFFLINE (red), DISARMED (green), or ARMED (amber). |
| **Conversation panel (main)** | Your messages on the right. AI replies on the left. Red alert messages when the aircraft needs atte |
| **Input box (bottom)** | Type here and press Enter, or click Send. |
| **RAW DATA button** | Shows the technical data stream. Not needed for normal operation. |

## How to talk to the AI

Speak to the AI the way you would speak to a knowledgeable co-pilot. Use whatever words feel natural.

*"Ready to fly"*
→ AI runs all pre-flight checks and reports anything needing attention.

*"Arm it"*
→ AI confirms once, then arms. Status badge turns amber.

*"It's drifting left"*
→ AI checks the attitude hold system, diagnoses the cause, and adjusts.

*"Hover here"*
→ AI engages GPS position hold.

*"Teach it to turn tighter"*
→ AI sets the aerobatics learning focus.

*"How much battery left?"*
→ AI reads voltage and tells you in plain language.

*"Land it"*
→ AI guides through the landing sequence.

*"We're done for today"*
→ AI saves session learning, then disarms.

◆ **NOTE** The AI watches the instruments all the time. If it sees a fault, low battery, or unsafe condition, it speaks up without waiting for you to ask.

## Safety confirmations

For arming, disarming in flight, emergency stop, and ceiling exploration, the AI asks you to confirm before acting. Ceiling exploration requires two separate confirmations due to airspace implications (Section 12).

■ **IMPORTANT** Emergency stop cuts all motors immediately. The aircraft drops. Only use this if a crash is certain and people are at risk.

# 5 Pre-Flight Checks

Before every flight, run a pre-flight check. You do not need to remember what to check — tell the AI you are ready.

> *"Ready to fly"*
> → AI checks: faults, battery voltage, attitude hold, hover mode, airspeed sensor, GPS lock, and rope health. Reports anything needing attention.

## What the AI checks

| | |
|---|---|
| **Faults** | Any errors from the previous flight or current power-up. |
| **Battery** | Must be above 21.5V. Below 20V: do not fly. |
| **Attitude hold** | The system that keeps the aircraft pointing where commanded. |
| **Hover mode** | The velocity-stick control used at low speed. |
| **Airspeed sensor** | The pitot tube that measures flying speed. |
| **GPS** | Position must be locked before autonomous features work. |
| **Rope health** | The servo hinge ropes — checked before aggressive flight. |

## If checks fail

The AI tells you what failed and what to do. Do not arm if any check fails. Common causes:

**Battery below 21.5V** — Charge the battery.

**GPS not locked** — Wait 60–90 seconds in the open, away from buildings.

**Fault from previous flight** — Ask the AI what the fault was — it explains in plain English.

**Airspeed sensor fault** — Check nothing is blocking the pitot tube at the nose.

## Arming

Once checks pass, tell the AI to arm. It confirms once, then arms. The status badge turns amber and shows ARMED.

■ **CAUTION** Stand clear of all eight wings once armed. Wings can move at any time.

◆ **NOTE** To disarm on the ground: "disarm" or "we're done". Do not disarm in the air unless instructed by the AI in a fault situation.

# 6 Normal Flight Operations

The MCO-1D has two flight modes — hover and fixed-wing cruise — and transitions between them automatically as airspeed changes.

## The two flight modes

| | |
|---|---|
| HOVER (below ~3 m/s) | Wings flap to maintain lift. Lever = altitude. Joystick = horizontal velocity — push to move, release to st |
| FIXED-WING (above ~5 m/s) | Flies like a conventional aircraft. Lever = thrust/braking. Joystick = bank, pitch, and yaw. Between 3 an |

## Taking off

Say "take off" or "let's go". The AI increases throttle until the aircraft lifts clear, then holds at a safe hover height. You can also take off manually by advancing the lever.

## In-flight assistance

During flight the AI monitors continuously. You can ask anything at any time:

- *"How high are we?"*
- *"What's the battery at?"*
- *"Is anything wrong?"*
- *"Why is it doing that?"*
- *"Hold position here"*
- *"Come back towards me"*

## Turn tightening

During banked turns at cruise speed, the aircraft automatically uses its most effective wing mechanisms to produce tighter turns. Harder stick at higher airspeeds produces progressively tighter turns. This is invisible to the pilot.

## Landing

Say "land" or "bring it down". For a steeper approach say "steep approach". For aggressive braking deceleration see Section 8.

◆ **NOTE** After touchdown wait for the aircraft to stop before approaching. Say "we're done" to save learning and disarm.

# 7 The Lever — Complete Guide

The throttle lever controls the aircraft's energy state. Forward = more power. Back = less power and progressively stronger braking. 50% (centre) holds altitude.

◆ **NOTE** The lever follows fixed-wing convention throughout. Forward is more power. This is the same as any fixed-wing aircraft you have flown.

| ZONE | LEVER | WHAT HAPPENS |
|---|---|---|
| THRUST | 0–5% | Maximum burst at low speed, or fast forward sprint at cruise speed. |
| SPRINT | 5–28% | High-speed sprint. Wings flapping fastest. Maximum thrust. |
| POWERED CRUISE | 28–45% | Normal cruise. Good efficiency. Accelerates to cruise speed. |
| ALTITUDE HOLD | 45–55% | Holds current altitude. 50% is the exact hold point. |
| GLIDE / BEST RANGE | 55–75% | Efficient descent and glide. Kestrel mode can engage here in good wind. |
| LIFT DUMP PARTIAL | 75–80% | Rear wings reduce lift — steeper approach, like partial spoilers. |
| LIFT DUMP FULL | 80–87% | Rear wings stop — steep approach, full spoilers equivalent. |
| SHUTTLE BRAKE PARTIAL | 87–93% | All wings begin braking progressively. |
| J-HOOK | 93–97% | Post-stall pitch-up to hover. Requires 20m+ altitude. See Section 8. |
| SHUTTLE BRAKE FULL | 97–100% | Maximum drag. Emergency deceleration (~2g). Auto-exits when slow. |

The pitch stick modifies brake intensity when the lever is above 87%. Pulling back adds more brake; pushing forward reduces it. Roll and yaw sticks remain fully active throughout all brake modes.

# 8 Braking and Deceleration

The MCO-1D uses aerodynamic braking modes triggered by lever position — no separate switch needed.

## Lift dump — approach control

Lever at 75–87% stops the rear wing pairs, reducing total lift without adding drag. This steepens the approach — useful for landing in a confined space. The effect builds gradually: 75% is a gentle steepen, 87% is maximum slope.

*"Steepen the approach"*
→ AI moves the lever into the lift dump zone and explains what the aircraft is doing.

## Shuttle brake — aerodynamic deceleration

At 87–97% lever, all wings pitch to high angles and slow their flapping, dramatically increasing drag. Like a shuttlecock facing the airflow. Deceleration builds smoothly. Full shuttle brake at 93–97% gives 1.5–2g deceleration.

◆ **NOTE** The shuttle brake exits automatically when airspeed drops below 2 m/s, transitioning smoothly into hover. No manual release needed.

## J-hook — post-stall pitch-up to hover

At 93–97% lever above 20 metres, a three-phase pitch-up engages: rapid nose-up, hold while decelerating, then transition to hover as speed drops. Birds use this mechanism to land on perches. Completes in under 3 seconds.

■ **CAUTION** J-hook requires minimum 20m altitude. Below this the system falls back to shuttle brake. The AI warns you if altitude is insufficient.

*"Stop fast / Emergency brake"*
→ AI checks altitude, explains modes, engages most appropriate.

*"J-hook"*
→ AI checks height above 20m, confirms, engages the post-stall pitch-up.

## Emergency stop

Emergency stop cuts all motor power instantly. The aircraft falls. Only for situations where a crash is already happening and people are in the flight path.

■ **IMPORTANT** Never use emergency stop as a landing method. Say "land" or "abort" for all normal situations.

# 9 How the Aircraft Learns

The MCO-1D improves its own performance continuously while flying. You do not need to do anything. Every flight hour makes the aircraft better.

## What is happening during flight

While in stable cruise, the aircraft continuously tests small variations in wing configuration — angles, timing, sweep, and amplitude. It measures performance, keeps better variants, discards worse ones. This is called discovery.

The aircraft also maintains an archive of the best configurations found for every combination of airspeed and wind condition encountered. Over many flights this archive fills in and the aircraft gets better at each specific condition separately.

A third system — the strategy genome — learns which wing mechanism combinations work best for each manoeuvre: rolls, climbs, braking, hover hold. It evolves like a population of strategies, the best ones breeding new variations.

## What you can do to help

Tell the AI what you want the aircraft to get better at. That is Mission Focus (Section 10). Beyond that, flying more is learning more — every battery cycle adds to the aircraft's knowledge.

*"That worked really well"*
→ AI records a positive observation. That configuration gets credit in the archive.

*"The hover feels wobbly in this crosswind"*
→ AI sets the learning focus to crosswind stability.

## Saving learning between flights

All learning is saved when you say "we're done" or "land and save". Stored on the Pi, survives power cycles. After many flights the aircraft carries knowledge from every session.

■ **CAUTION** Never disconnect the battery without saving. Say "save learning" first if needed. A missed save loses only that session — all previous flights remain intact.

# 10 Mission Focus

Mission Focus tells the aircraft what to get good at. The learning system concentrates on the specific condition or capability you describe.

## Setting a focus

Use any words that describe what you need:

| WHEN YOU SAY… | WHAT IT DOES |
|---|---|
| *"It's gusty today"* | Optimises for stability in wind and turbulence. |
| *"Hold position better"* | Concentrates on GPS position hold accuracy. |
| *"Smooth flight / camera payload"* | Minimises vibration and attitude variation. |
| *"Landing in a crosswind"* | Optimises crosswind approach stability. |
| *"Stay up longer / battery"* | Minimises power for maximum endurance. |
| *"Let's see how fast"* | Optimises for maximum cruise speed. |
| *"SAR mission"* | Optimises for SAR hover and confined-space ops. |
| *"Really stormy"* | Loads storm-resilience configurations. |
| *"Tight aerobatic turns"* | Optimises roll authority and turn tightening. |
| *"Good updraft here"* | Tries to engage kestrel soaring mode. |
| *"Back to normal"* | Clears focus, returns to general optimisation. |

## How long does focus take to work?

The learning system starts immediately. Some improvements appear within the same flight. Larger improvements accumulate over multiple flights on the same focus. Some focuses (SAR hover, storm resilience, kestrel, endurance) persist between flights.

*"How is the learning going?"*
→ AI summarises configurations tested, best performers, and current focus effect.

*"What has it improved at?"*
→ AI describes the most significant performance gains since learning began.

# 11 Autonomous Learning Sorties

The Autonomous Learning Sortie System (ALSS) lets the aircraft fly, learn, and repeat with minimal human involvement. Define the flying area, start the programme, swap batteries.

## What happens during a session

The aircraft takes off, works through a planned sequence of manoeuvres, returns to the landing pad, and waits. You swap the battery and arm again. The next flight adapts based on what the previous one learned — spending more time on low-scoring manoeuvres, advancing as competence builds.

## The three learning classes

| | |
|---|---|
| CLASS A (first 5 sorties) | Orientation. Hover at three heights, gentle cruise circuits, gentle rolls, climbs and descents. Establishe |
| CLASS B (sorties 6–20) | Exploration. Efficiency cruise, moderate banked turns, lift dump approach, hard climbs, clean glide. Fi |
| CLASS C (sortie 21+) | Performance. Full roll authority, VIFF turns, shuttle brake, J-hook, sprint, lag-roll. Only unlocks after C |

## Setting up an ALSS session

1. **Define the flying area**

   Tell the AI the GPS centre position and size. The aircraft will not fly outside it.

   Example: "Set the practice area to 51.5074, -0.1278, 100 metres radius, 50 metres high"

2. **Start the programme**

   Say "start the practice flights" or "begin autonomous learning". The AI confirms the area and begins.

3. **Swap batteries when prompted**

   When the aircraft lands and says "swap battery and re-arm", swap the battery and say "arm" or "start the next flight". The next sortie begins automatically.

   All learning is saved automatically between sorties.

4. **Monitor progress**

   Ask "how is the practice going?" or "show me the scores" at any time.

5. **End the session**

   Say "stop the practice" or "end the session". The AI completes the current sortie, lands, saves, and disarms.

■ **CAUTION** A human safety observer should be present, especially for first Class C sorties. To abort immediately: say "abort" or "land now".

# 12 Ceiling Exploration

■ **IMPORTANT** Ceiling exploration may reach altitudes requiring notification or authorisation under aviation regulations. Obtain all necessary permissions before proceeding.

## What the aircraft is looking for

As altitude increases, air gets thinner and the aircraft works harder — wings flap faster with wider strokes. At some altitude the wings reach hardware limits. That is the ceiling. The aircraft approaches it very slowly, holds each new altitude for 60 seconds to test stability, and records the best configuration at each height.

The aircraft also watches for LEV collapse — where the vortex generating wing lift breaks down in thin air. If this risk rises above a threshold the aircraft descends automatically.

## The two-step authorisation procedure

Two separate confirmations are required. This is intentional — it prevents accidental engagement of a procedure with airspace implications.

**1** **Complete the checklist (prompted by the AI)**

When you say "arm ceiling exploration" the AI displays a checklist:

- ■ Airspace authorisation obtained for this altitude and area
- ■ NOTAM filed or authority notified as required
- ■ Chase observer in position with clear line of sight
- ■ Wind below 5 m/s at the surface
- ■ Battery fully charged
- ■ Aircraft in stable cruise with discovery enabled

**2** **Confirm to begin (within 60 seconds)**

Say "confirm ceiling exploration" within 60 seconds of the checklist. If more than 60 seconds pass, the arm expires and you must start again.

This window is intentional — you cannot arm and walk away.

## During ceiling exploration

The aircraft climbs at 0.15 metres per second — very slowly by design. The probe requires patience: 60 seconds at each altitude before climbing further. A full probe can take 30–60 minutes.

*"How high are we?"*

→ AI gives current altitude and ceiling phase.

*"Abort the ceiling test"*

→ AI descends immediately and ends the probe.

# 13 After Flight

Proper after-flight procedure preserves the aircraft's learning and leaves hardware in a safe state for next time.

## Standard shutdown sequence

1. **"We're done" or "land and save"**
   The AI saves all session learning, then disarms. This single phrase does everything needed for a normal shutdown.

2. **Wait for the disarmed confirmation**
   Status badge returns to green (DISARMED). Do not approach the aircraft until the badge shows DISARMED.

3. **Disconnect the battery**
   Only after the AI confirms the aircraft is disarmed and saved.

4. **Close the ground station**
   Press Ctrl+C in the terminal. Stops both the aircraft link and AI link cleanly.

## If you need to stop quickly

Say "save learning" before disconnecting. Saves everything immediately.

> ◆ **NOTE** If you disconnected without saving: only that session's discoveries are lost. All previous flights remain intact on the Pi.

## Between sessions

The aircraft's knowledge accumulates automatically. Each new flight adds to the archive. At the next session the aircraft starts with everything it has ever learned.

# 14 Troubleshooting

Start with the AI — it can diagnose many faults by reading the aircraft's data directly.

### Interface shows OFFLINE

Launcher not running, or aircraft not powered or reachable. Check: (1) launcher running, (2) aircraft light solid green, (3) laptop on same network, (4) ping 192.168.7.1 gets replies.

### Pre-flight fails on battery

Charge the battery. Must be above 21.5V. Below 20V: do not fly.

### Pre-flight fails on GPS

Move to open area away from buildings. Wait 60–90 seconds.

### Pre-flight shows a fault

Say "what is the fault?" — AI explains in plain English.

### Aircraft drifts in hover

Say "it's drifting". AI checks attitude hold and GPS, adjusts. "Hold position" for active position correction.

### Aircraft sluggish in turns

Say "the turns feel sluggish". AI sets aerobatics focus and begins optimising. Improves over a few minutes of flight.

### AI doesn't understand me

Rephrase more simply. Say "no, I meant…" and it tries again.

### Not responding to lever input

Say "what's wrong" immediately. "Emergency stop" only if collision with a person is imminent.

### Lost connection during flight

Aircraft returns to take-off point and lands autonomously. Do not approach until landed and powered down.

### Ceiling probe stopped unexpectedly

Aircraft detected a flight limit and descended safely. Ask "why did it stop?" — AI explains what limit was reached.

# Appendix A Useful Phrases

The AI understands many phrasings for each intent. Use whatever feels natural.

## Starting and arming

*"Ready to fly"*
→ Runs all pre-flight checks.

*"Arm it / Let's arm"*
→ Arms after confirmation.

*"Disarm / Shut down"*
→ Disarms on the ground.

## Flight control

*"Take off / Let's go"*
→ Begins takeoff sequence.

*"Land / Bring it down"*
→ Begins landing sequence.

*"Hold position / Stay here"*
→ Engages GPS position hold.

*"Come back / Return"*
→ Returns to launch point.

*"Steep approach / Short landing"*
→ Engages lift dump.

*"Stop fast"*
→ Engages shuttle brake.

*"J-hook / Hard stop"*
→ Engages J-hook if altitude allows.

## Monitoring

*"How high are we?"*
→ Reads altitude.

*"How much battery?"*
→ Reads voltage and estimates time remaining.

*"Is anything wrong?"*
→ Reads fault status.

*"Why is it doing that?"*
→ Reads telemetry and explains behaviour.

## Learning and focus

*"It's gusty / Handle the wind"*
→ Sets gust tolerance focus.

*"Stay up longer"*
→ Sets endurance focus.

*"Turn tighter / More agile"*
→ Sets aerobatics focus.

*"SAR mode / Rescue"*
→ Sets SAR hover focus.

*"That worked well"*
→ Records a positive performance note.

*"Back to normal"*
→ Clears the current focus.

## Autonomous sorties

*"Start practice flights"*
→ Starts ALSS.

*"How are the sorties going?"*
→ Shows ALSS progress report.

*"Stop the practice"*
→ Ends ALSS after current sortie.

*"Abort / Land now"*
→ Immediately ends sortie and lands.

## Ceiling exploration

*"Arm ceiling exploration"*
→ Begins two-step authorisation.

*"Confirm ceiling exploration"*
→ Step 2 (within 60 seconds).

*"Abort the ceiling test"*
→ Descends immediately.

## Ending the session

*"We're done / End of day"*
→ Saves learning and disarms.

*"Save the learning"*
→ Saves immediately without disarming.

*"Shut everything down"*
→ Saves, disarms, prompts to close launcher.

# Appendix B What the Aircraft Says

The AI speaks on behalf of the aircraft. This appendix explains common messages.

**"Alert: fault detected"**
Hardware or software fault detected. Ask "what is the fault?" immediately.

**"Alert: battery low (X.Xv)"**
Battery dropping. Plan to land soon. Below 20V: land immediately.

**"Alert: wing stall risk at this altitude"**
During ceiling exploration: wing aerodynamics becoming unstable. Descending automatically.

**"All checks pass. Ready to arm."**
Pre-flight checks completed normally. Safe to arm.

**"Arming now."**
The aircraft is arming. Stand clear of the wings.

**"Discovery is running."**
Learning system actively testing configurations. Normal cruise behaviour.

**"Ceiling exploration — Step 1 of 2: Armed."**
Ceiling probe authorisation started. 60 seconds to confirm.

**"Ceiling probe: saturation at Xm AGL."**
Aircraft reached hardware limit at this altitude. Holding and running discovery loop.

**"ALSS Sortie N complete. Swap battery and re-arm to continue."**
Autonomous learning sortie finished. Swap battery, arm, next sortie begins.

**"Session saved. Safe to disconnect."**
All learning saved. Safe to disconnect the battery.

# MCO-1D Operator Manual

Revision 20 — August 2026

For hardware, assembly, commissioning, and test card procedures, refer to the MCO-1D Commissioning Manual.